

Question	Answer	Marks
9(a)	temperature inversely proportional to wavelength	M1
	temperature is thermodynamic temperature of surface of star and wavelength is the wavelength at which maximum emission rate from star occurs	A1
9(b)	<i>Any three points from:</i> • (surface) temperature of star X = 7000 K or star X has a higher temperature than the Sun • star X has a higher luminosity than the Sun • luminosity of star X = 2.7×10^{27} W • radius of star X = 1.3×10^9 m	B3
9(c)	light (from star X) is redshifted	B1
	wavelength of peak emission rate would be greater (using observed data)	B1

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